2026 MULTIMEDIA WIRELESS NETWORKS (ET5907701)

Analysis of IEEE 802.11e for QoS Support in WLANs

Assignment 3: Literature Review & Revised Project Proposal

Presenter
Abel Ledrappier (E11502008)

Instructor
Prof. Ray-Guang Cheng

Reference Paper
Mangold et al., IEEE Wireless Comm. 2003

1. PROBLEM & MOTIVATION: LEGACY 802.11 LIMITATIONS

PROBLEM STATEMENT

Distributed Coordination Function (DCF) Flaws

- **Homogeneous Channel Access:** All stations contend with identical parameters (DIFS, CWmin = 15, CWmax = 1023).
- **Zero Differentiation:** CSMA/CA treats real-time voice and background file transfers identically on the air interface.
- **Quality Degradation:** Under moderate-to-heavy traffic loads, queuing delays and collision frequencies surge unpredictably.

Point Coordination Function (PCF) Flaws

- **Unpredictable Beacon Delay:** Ongoing DCF transmissions across TBTT can delay the CFP beacon by up to **4.9 ms** in 802.11a.
- **Uncontrolled Transmission Time:** Polled stations can transmit arbitrary MSDUs up to 2304 bytes; AP cannot bound polled durations.
- **Hardware Adoption Failure:** Inefficiencies and lack of deterministic bounds prevented real-world commercial hardware deployment.

Core Motivation: Real-time multimedia streaming (VoIP, interactive video) mandates bounded delay, minimal jitter, and guaranteed throughput—necessitating a protocol-level MAC overhaul.

2. IEEE 802.11E QOS MECHANISMS: EDCA & HCCA

4 Access Categories (ACs)

- **AC_VO:** Voice (highest priority).
- **AC_VI:** Video.
- **AC_BE:** Best Effort.
- **AC_BK:** Background (lowest priority).
- Mapped from 8 user priorities (IEEE 802.1D) into 4 parallel backoff entities per station.

EDCA Differentiation Levers

- **AIFS[AC]:** Arbitration Interframe Space replaces DIFS (AIFS = SIFS + AIFSN × aSlotTime).
- **Contention Window:** Dedicated [CWmin, CWmax] ranges per AC.
- **Virtual Collision Handler:** Resolves internal queue collisions locally in favor of higher priority.

Transmission Opportunity (TXOP)

- **Bounded Channel Holding:** Stations win time intervals (TXOP), not single packet transmissions.
- **EDCA-TXOP vs HCCA-TXOP:** Contention-based vs HC polled allocation.
- **Continuation & Bursting:** Multiple MSDUs transmitted consecutively separated by SIFS up to TXOPlimit.

Key Architectural Shift: Prioritization is achieved probabilistically in contention periods (EDCA) and deterministically through polling by the Hybrid Coordinator (HCCA).

3. SYSTEM MODEL & EXPERIMENTAL SETUP

Network & PHY Layer Configuration

- **Standard:** IEEE 802.11a at 5 GHz with OFDM.
- **PHY Data Rate:** 24 Mb/s (data frames), 6 Mb/s (control frames).
- **Timing:** Slot = 9 µs, SIFS = 16 µs, PIFS = 25 µs, DIFS = 34 µs.
- **Radio Channel:** Stochastic radio error model, all nodes in direct transmission range (no hidden nodes).
- **Mechanisms:** RTS/CTS disabled, packet fragmentation disabled.

Traffic Inputs & Performance Metrics

- **Traffic Model:** Downlink from HC/AP; 512-byte MSDU payloads with negative-exponential inter-arrival times.
- **Streams:** 4 concurrent streams per station (one per AC, 250 kb/s baseline each).
- **Throughput Metric:** MAC-layer saturation throughput (Mb/s) as a function of offered load and node count.
- **Delay Metric:** Complementary CDF (CCDF) of delivery delay evaluated via Limited-Relative-Error algorithm (≤ 5% error).

Access Category	Designated Traffic	AIFSN	CWmin (slots)	CWmax (slots)	TXOPlimit
AC_VO	Voice	2	7	15	1 MSDU (in baseline evaluation)
AC_VI	Video	2	15	31	1 MSDU
AC_BE	Best Effort	3	31	1023	1 MSDU
AC_BK	Background	7	31	1023	1 MSDU

4. KEY RESULT 1: SERVICE DIFFERENTIATION UNDER LOAD (FIGURE 8)

Throughput Saturation Analysis

- **Linear Growth Phase (< 3 Mb/s per AC):** Channel is uncongested; all four access categories satisfy 100% of their offered traffic.
- **Preemption at Heavy Load (> 4 Mb/s):** As total load approaches channel saturation (~14 Mb/s MAC throughput), priority separation begins immediately.
- **Starvation of AC_BK & AC_BE:** Background traffic drops to near zero at 4 Mb/s load; Best Effort collapses at 9 Mb/s.
- **Protection of AC_VO & AC_VI:** Voice reaches ~13 Mb/s saturated throughput unimpeded; Video secures remaining capacity (~1.5 Mb/s).

Underlying Protocol Mechanics

- **Shorter Arbitration Delay:** AIFSN = 2 for VO/VI allows them to start backoff countdown 1 to 5 slots earlier than BE/BK.
- **Rapid Window Clearance:** CWmin = 7 gives AC_VO an average backoff of only 3.5 slots vs 15.5 slots for BE/BK.
- **Internal Preemption:** Virtual collision handling automatically defers lower queues without wasting physical airtime.

Takeaway for A4 Reproduction: EDCA effectively provides tiered QoS differentiation during overload, making Figure 8 the ideal baseline for ns-3 validation.

5. KEY RESULT 2: CONTENTION COLLAPSE IN DENSE SCENARIOS

Figure 9: Scalability Breakdown

- **Scenario:** Scaling contending stations from 1 to 16, each offering 1 Mb/s across 4 ACs.
- **Abrupt Collapse:** Beyond 5 contending stations, aggregate AC_VO throughput drops sharply from 4.7 Mb/s down to < 0.5 Mb/s at 16 stations.
- **Root Cause:** CWmin = 7 is excessively small for large node sets; multiple voice entities repeatedly draw identical random timers, resulting in fatal collision loops.

Figure 10: HCCA Delay vs Overlapping BSS

- **Isolated QBSS:** Polled HCCA access strictly bounds MSDU delivery delay within the configured TXOPlimit threshold.
- **Overlapping QBSS (OBSS) Failure:** When two neighboring HCs operate on the same channel without synchronization, poll frames collide, destroying QoS delay guarantees.
- **Design Conclusion:** Static EDCA parameters require dynamic runtime adaptation, while HCCA requires inter-BSS coordination.

Core Insight: Static EDCA parameters inherently fail in dense multi-station environments—providing the direct motivation for our Final Project modification.

6. BASELINE REPRODUCTION PLAN IN NS-3 (ASSIGNMENT 4)

REPRODUCTION STRATEGY

1. Topology & PHY Setup

- 1 AP (Hybrid Coordinator) and 3 associated STAs in close proximity.
- `WifiHelper` configured with `802.11a`, ConstantRateManager at `OfdmRate24Mbps`.
- Control frames pinned at `OfdmRate6Mbps`.

2. EDCA Parameter Mapping

- Use native `ns3::QosTxop` objects.
- Set `Aifsn`, `MinCw`, and `MaxCw` per AC according to Table 1.
- Configure `TxopLimit = MicroSeconds(0)` to restrict transmissions to 1 MSDU per winning contention.

3. Traffic & Trace Harness

- `OnOffApplication` / `PacketSocketClient` generating 512-byte UDP packets with exponential off-times.
- Sweep load from 0 to 30 Mb/s per AC across multiple independent RNG seeds.
- Extract throughput via `FlowMonitor` and PCAP tracing.

Deliverable for A4: Replicate Figure 8 throughput curves directly inside ns-3, plotting simulation outputs against Mangold's published data points.

7. REVISED PROPOSAL: TRACK SHIFT FROM ADVANCED (NTN) TO BASIC

PROPOSAL REVISION

Initial A1 Proposal: 5G Non-Terrestrial Networks

- **Initial Concept:** QoS modeling and Random Access for LEO satellite constellations under 3GPP NTN specifications.
- **Technical Roadblocks:**
  - Out-of-tree simulator forks (e.g., custom satellite modules) suffered critical build issues with recent ns-3 toolchains.
  - High model complexity (orbital dynamics, Doppler shift, massive propagation delays) diverted focus from core MAC QoS mechanisms.
  - Lack of a clean, standardized IEEE reference paper with easily replicable baseline curves for A4.

Strategic Pivot to Basic Path (IEEE 802.11e)

- **Native ns-3 Integration:** EDCA and 802.11e are fully supported within upstream `src/wifi`, guaranteeing reproducible experiments.
- **Direct Alignment with Course Outline:** Perfectly synchronizes with Weeks 4–6 lectures on WiFi MAC and EDCA fundamentals.
- **Guaranteed Feasibility:** Ensures a seamless progression from Literature Review (A3) → Baseline Reproduction (A4) → Final Project Modification.

Decision Rationale: Shifting to the Basic Path trades external software friction for rigorous protocol evaluation and algorithmic innovation.

8. PROPOSED MODIFICATION: ADAPTIVE CONTENTION WINDOW (A-EDCA)

FINAL PROJECT VISION

Targeted Problem: Voice Collapse Under Density

- As demonstrated in **Figure 9**, static parameters (CWmin = 7) cause high-priority voice traffic to collapse once contending stations exceed 5.
- In dense enterprise/campus networks, static parameters render EDCA unable to sustain VoIP call admissions.

Proposed Solution & Implementation Concept

- **Collision-Aware Dynamic Tuning:** AP monitors channel busy fraction and transmission failure rates.
- **Dynamic Beacon Parameter Updates:** HC updates CWmin[AC_VO] dynamically via Beacon frame information elements as node density grows.
- **Station-Side Adaptation:** Alternatively, explore localized multiplicative backoff adaptations to resolve heavy internal contention.

Evaluation Dimension	Baseline EDCA (A4)	Modified Adaptive EDCA (Final Project)
Parameter Policy	Static CWmin = 7, CWmax = 15	Dynamic CWmin(n) scaled by observed collision rate
Performance at 16 Stations	Severe throughput collapse (< 0.5 Mb/s)	Sustained saturation throughput (> 3.0 Mb/s)
Delay & Jitter	High jitter due to repeated collision retries	Bounded delay distribution for voice streams

9. CONCLUSION & PROJECT ROADMAP

MILESTONES

Assignment 3 (Completed)

- Literature review of foundational 802.11e paper.
- System model, parameters, and key results extracted.
- Proposal formally pivoted to Basic Track.
- Study note and rehearsal feedback published in GitHub.

Assignment 4 (Next Step)

- Implement ns-3 simulation script matching Section 3 parameters.
- Run parameter sweeps across 0–30 Mb/s offered load.
- Reproduce Figure 8 throughput curves.
- Document discrepancies and calibration factors.

Final Project (Weeks 15–16)

- Implement dynamic contention window adaptation algorithm.
- Benchmark against A4 baseline in dense station scenarios (Figure 9 conditions).
- Deliver comprehensive project report and final presentation.

Thank you for your attention. Questions & Answers.